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Tutorial Course for Intro to Sensor-Integrated Systems

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2026

1. A pressure measurement system consists of a sensor, amplifier, and display. Table below gives the model equations, mean values, and standard deviations for each element in the system. All probability distributions are normal.

	Sensor	Amplifier	Display
Model equations	$R_p = K_1 \exp\left(\frac{Y}{p}\right)$	$V_A = K_2 \left(\frac{2}{1 + \frac{2.5}{R_p}} \right) - b_1$	$P_M = K_3 V_A + b_2$
Mean values	$\overline{K_1} = 4 \text{ k}\Omega$ $\overline{\gamma} = 400 \text{ kPa}$	$\overline{K_2} = 2 \text{ V}$ $\overline{b_1} = 0.5 \text{ V}$	$\overline{K_3} = 80 \text{ kPa/V}$ $\overline{b_2} = -50 \text{ kPa}$
Std deviations	$\sigma_{K_1} = 0.3 \times 10^{-3}$ $\sigma_{\gamma} = 2$	$\sigma_{K_2} = 0.05$ $\sigma_{b_1} = 0.02$	$\sigma_{K_3} = 2$ $\sigma_{b_2} = 1.5$

- Please calculate the mean of the error probability density function for a true input pressure of 200 kPa. (5 points)
- Please calculate the standard deviation of the error probability density function for a true input pressure of 50 kPa. (10 points)

Solution

Part a) Mean of the error probability density function at $P_{\text{true}} = 200 \text{ kPa}$

The mean error ($\overline{E}$) is the difference between the mean measured pressure ($\overline{P_M}$) and the true input pressure (P_{true}):

$$\overline{E} = \overline{P_M} - P_{\text{true}}$$

1. Sensor Mean Output ($\overline{R_P}$):

Substitute $P = 200 \text{ kPa}$ and the mean values into the model:

$$\overline{R_P} = \overline{K_1} \exp\left(\frac{\overline{\gamma}}{P}\right)$$

$$\overline{R_P} = 4 \exp\left(\frac{400}{200}\right) = 4 \exp(2) \approx 29.556 \text{ k}\Omega$$

2. Amplifier Mean Output ($\overline{V_A}$):

$$\overline{V_A} = \overline{K_2} \left(\frac{2}{1 + \frac{2.5}{\overline{R_P}}} \right) - \overline{b_1}$$

$$\overline{V_A} = 2 \left(\frac{2}{1 + \frac{2.5}{29.556}} \right) - 0.5 \approx 3.188 \text{ V}$$

3. Display Mean Output ($\overline{P_M}$):

$$\overline{P_M} = \overline{K_3} \overline{V_A} + \overline{b_2}$$

$$\overline{P_M} = 80(3.188) - 50 = 205.04 \text{ kPa}$$

4. Mean Error Calculation:

$$\overline{E} = 205.04 - 200 = 5.04 \text{ kPa}$$

Solution

Part b) Standard deviation of the error probability density function at $P_{\text{true}} = 50$ kPa

The standard deviation of the error equals the standard deviation of the measured output ($\sigma_E = \sigma_{P_M}$). We use the uncertainty propagation formula (variance synthesis):

$$\sigma_y^2 = \sum \left(\frac{\partial y}{\partial x_i} \right)^2 \sigma_{x_i}^2$$

1. Calculate Means at $P = 50$ kPa:

- $\overline{R_P} = 4 \exp(400/50) = 4 \exp(8) \approx 11923.83 \text{ k}\Omega$
- $\overline{V_A} = 2 \left(\frac{2}{1+2.5/11923.83} \right) - 0.5 \approx 3.499 \text{ V}$
- $\overline{P_M} = 80(3.499) - 50 \approx 229.93 \text{ kPa}$

2. Variance of Sensor ($\sigma_{R_P}^2$):

$$\sigma_{R_P}^2 = \left(\frac{\partial R_P}{\partial K_1} \right)^2 \sigma_{K_1}^2 + \left(\frac{\partial R_P}{\partial \gamma} \right)^2 \sigma_{\gamma}^2$$

- $\frac{\partial R_P}{\partial K_1} = \exp(8) \approx 2980.96$
- $\frac{\partial R_P}{\partial \gamma} = \frac{4}{50} \exp(8) \approx 238.48$

$$\sigma_{R_P}^2 = (2980.96 \times 0.3 \times 10^{-3})^2 + (238.48 \times 2)^2 \approx 0.8 + 227484.4 \approx 227485.2$$

Solution

3. Variance of Amplifier ($\sigma_{V_A}^2$):

$$\sigma_{V_A}^2 = \left(\frac{\partial V_A}{\partial K_2}\right)^2 \sigma_{K_2}^2 + \left(\frac{\partial V_A}{\partial b_1}\right)^2 \sigma_{b_1}^2 + \left(\frac{\partial V_A}{\partial R_P}\right)^2 \sigma_{R_P}^2$$

- $\frac{\partial V_A}{\partial K_2} \approx 1.9996$
- $\frac{\partial V_A}{\partial b_1} = -1$
- The propagated variance from R_P is negligible due to the small partial derivative.

$$\sigma_{V_A}^2 \approx (1.9996 \times 0.05)^2 + (-1 \times 0.02)^2 \approx 0.0100 + 0.0004 = 0.0104 \text{ V}^2$$

4. Variance of Display ($\sigma_{P_M}^2$) and Final Standard Deviation:

$$\sigma_{P_M}^2 = \left(\frac{\partial P_M}{\partial K_3}\right)^2 \sigma_{K_3}^2 + \left(\frac{\partial P_M}{\partial b_2}\right)^2 \sigma_{b_2}^2 + \left(\frac{\partial P_M}{\partial V_A}\right)^2 \sigma_{V_A}^2$$

- $\frac{\partial P_M}{\partial K_3} = V_A = 3.499$
- $\frac{\partial P_M}{\partial b_2} = 1$
- $\frac{\partial P_M}{\partial V_A} = K_3 = 80$

$$\sigma_{P_M}^2 = (3.499 \times 2)^2 + (1 \times 1.5)^2 + (80)^2 \times 0.0104$$

Take the square root to find the standard deviation:

$$\sigma_{P_M}^2 \approx 48.97 + 2.25 + 66.56 = 117.78 \text{ kPa}^2$$

$$\sigma_E = \sqrt{117.78} \approx 10.85 \text{ kPa}$$

2. A temperature measurement system consists of linear elements and has an overall steady-state sensitivity of unity. The system dynamics are determined by a first-order lag (inertia) of the sensing element.

At time $t=0$, the sensor is suddenly transferred from ambient air at 25°C to an ice-water mixture at 0°C . After 90 seconds, the sensor is transferred to a constant-temperature oil bath at 60°C . Using the sensor data below, analyze and answer the following questions.

$$\text{Mass } m = 4 \times 10^{-2} \text{ kg}$$

$$\text{Surface area } A = 8 \times 10^{-4} \text{ m}^2$$

$$\text{Specific heat capacity } c = 0.25 \text{ J/(kg} \cdot ^{\circ}\text{C)}$$

$$\text{Heat transfer coefficient for air } h_{air} = 0.15 \text{ W/(m}^2 \cdot ^{\circ}\text{C)}$$

$$\text{Heat transfer coefficient for ice-water mixture } h_{ice} = 1.2 \text{ W/(m}^2 \cdot ^{\circ}\text{C)}$$

$$\text{Heat transfer coefficient for oil bath } h_{oil} = 0.8 \text{ W/(m}^2 \cdot ^{\circ}\text{C)}$$

- a) Please calculate the time constants of the sensor in the ice-water mixture and in the oil bath. (5 points)
- b) Consider only the ice-water and oil bath stages, write the temperature response equations for the sensor over the entire process (0–90 seconds in ice-water, after 90 seconds in oil bath). (5 points)
- c) Assume the oil bath temperature is suddenly increased to 80°C at $t = 120 \text{ s}$. Estimate the sensor readings at $t = 140 \text{ s}$. (5 points)

Solution

Part a) Time constants in the ice-water mixture and oil bath

The time constant τ for a first-order thermal sensor is calculated as:

$$\tau = \frac{mc}{hA}$$

First, calculate the thermal capacity (mc), which is constant:

$$mc = (4 \times 10^{-2} \text{ kg}) \times (0.25 \text{ J}/(\text{kg} \cdot ^\circ \text{C})) = 0.01 \text{ J}/^\circ \text{C}$$

1. Time constant in ice-water mixture (τ_{ice}):

$$\tau_{\text{ice}} = \frac{0.01}{1.2 \times (8 \times 10^{-4})} = \frac{0.01}{9.6 \times 10^{-4}} \approx 10.42 \text{ s}$$

2. Time constant in oil bath (τ_{oil}):

$$\tau_{\text{oil}} = \frac{0.01}{0.8 \times (8 \times 10^{-4})} = \frac{0.01}{6.4 \times 10^{-4}} = 15.625 \text{ s}$$

Solution

Part b) Temperature response equations (0–90 seconds and after 90 seconds)

The standard step response equation for a first-order system is:

$$T(t) = T_{\infty} + [T(t_0) - T_{\infty}]e^{-\frac{t-t_0}{\tau}}$$

1. Ice-water stage ($0 \leq t \leq 90$ s):

- Initial temp $T(0) = 25^{\circ}\text{C}$, Ambient temp $T_{\infty} = 0^{\circ}\text{C}$, $\tau_{\text{ice}} \approx 10.42$ s

$$T(t) = 0 + (25 - 0)e^{-\frac{t-0}{10.42}}$$

$$T(t) = 25e^{-\frac{t}{10.42}} \quad (0 \leq t \leq 90 \text{ s})$$

2. Oil bath stage ($t > 90$ s):

- At $t = 90$ s, the sensor has reached steady-state in ice water: $T(90) \approx 0^{\circ}\text{C}$.

- Ambient temp $T_{\infty} = 60^{\circ}\text{C}$, $\tau_{\text{oil}} = 15.625$ s

$$T(t) = 60 + (0 - 60)e^{-\frac{t-90}{15.625}}$$

$$T(t) = 60 - 60e^{-\frac{t-90}{15.625}} \quad (t > 90 \text{ s})$$

Solution

Part c) Estimate the sensor reading at $t = 140$ s

Step 1: Calculate the actual temperature at $t = 120$ s (before the sudden increase).

Using the oil bath equation from part b:

$$T(120) = 60 - 60e^{-\frac{120-90}{15.625}} = 60 - 60e^{-\frac{30}{15.625}} = 60 - 60e^{-1.92}$$

$$T(120) \approx 60 - 60(0.1466) = 60 - 8.796 = 51.204^{\circ}\text{C}$$

Step 2: Establish the new equation and calculate $T(140)$.

o Initial time $t_0 = 120$ s, Initial temp $T(120) = 51.204^{\circ}\text{C}$

o New ambient temp $T_{\infty} = 80^{\circ}\text{C}$, $\tau_{\text{oil}} = 15.625$ s

$$T(t) = 80 + (51.204 - 80)e^{-\frac{t-120}{15.625}} = 80 - 28.796e^{-\frac{t-120}{15.625}}$$

Substitute $t = 140$ s:

$$T(140) = 80 - 28.796e^{-\frac{140-120}{15.625}} = 80 - 28.796e^{-\frac{20}{15.625}}$$

$$T(140) = 80 - 28.796e^{-1.28} \approx 80 - 28.796(0.2780) \approx 80 - 8.005 = 71.995^{\circ}\text{C}$$

3. A batch of 80 identical smart temperature sensors were tested over a 9-month period (assume 1 month = 30 days). During this time, 16 failures were recorded, and the individual down times in hours for each failure were:

4, 6, 7, 5, 4, 8, 5, 6, 3, 7, 4, 8, 6, 7, 3, 5.

- a) Please calculate the mean time between failures. (5 points)
- b) Please calculate the mean failure rate (how many failures/year). (5 points)
- c) Assuming constant failure rate, compute the reliability of a 2-in-serial sensor configuration over a period of 2 months. (5 points)
- d) The sensors are to be used in a safety-critical monitoring system. Each monitoring point uses 3 sensors in parallel. Each monitoring point is connected in series with a signal conditioning module (mean failure rate: 0.8 failures/year) and a data acquisition module (mean failure rate: 0.25 failures/year). Please calculate the overall system reliability over 1 year. (5 points)

Solution

Part a) Calculate the mean time between failures (MTBF)

MTBF is defined as the total actual operating time divided by the total number of failures:

$$\text{MTBF} = \frac{\text{Total Operating Time}}{\text{Total Number of Failures}}$$

1. Calculate Total Possible Time:

- 80 sensors tested for 9 months (assuming 30 days/month, 24 hours/day).

$$T_{\text{total_possible}} = 80 \times 9 \times 30 \times 24 = 518,400 \text{ hours}$$

2. Calculate Total Down Time:

Sum of the individual down times for the 16 failures:

$$T_d = 4 + 6 + 7 + 5 + 4 + 8 + 5 + 6 + 3 + 7 + 4 + 8 + 6 + 7 + 3 + 5 = 88 \text{ hours}$$

3. Calculate Total Operating Time and MTBF:

$$T_{\text{operating}} = 518,400 - 88 = 518,312 \text{ hours}$$

$$\text{MTBF} = \frac{518,312}{16} = 32,394.5 \text{ hours}$$

Part b) Calculate the mean failure rate (failures/year)

The mean failure rate (λ) is the reciprocal of the MTBF.

$$\lambda_{\text{hourly}} = \frac{1}{\text{MTBF}} = \frac{1}{32,394.5} \approx 3.0869 \times 10^{-5} \text{ failures/hour}$$

Convert this to a yearly rate (assuming 1 standard year = $365 \times 24 = 8760$ hours):

$$\lambda_{\text{yearly}} = \lambda_{\text{hourly}} \times 8760$$

$$\lambda_{\text{yearly}} \approx 3.0869 \times 10^{-5} \times 8760 \approx 0.2704 \text{ failures/year}$$

Solution

Part c) Reliability of a 2-in-serial configuration over 2 months

Assuming a constant failure rate, reliability follows an exponential distribution: $R(t) = e^{-\lambda t}$.

Convert the time period to years: $t = 2 \text{ months} = 1/6 \text{ year}$.

1. Reliability of a single sensor:

$$R_{\text{single}} = e^{-\lambda t} = e^{-0.2704 \times (1/6)} \approx e^{-0.045067} \approx 0.9559$$

2. Reliability of the 2-in-serial system:

For a series system, $R_{\text{serial}} = R_1 \times R_2$.

$$R_{\text{serial}} = (R_{\text{single}})^2 \approx (0.9559)^2 \approx 0.9138$$

Part d) Overall system reliability over 1 year

The overall system consists of three main modules in series: a 3-sensor parallel module, a signal conditioning module, and a data acquisition module. Operating time $t = 1 \text{ year}$.

1. Reliability of the 3-sensor parallel module (R_{parallel}):

- Single sensor reliability over 1 year: $R_s = e^{-0.2704 \times 1} \approx 0.7631$
- Single sensor failure probability: $F_s = 1 - 0.7631 = 0.2369$
- 3 sensors failing simultaneously: $F_{\text{parallel}} = (F_s)^3 = (0.2369)^3 \approx 0.0133$
- Parallel module reliability: $R_{\text{parallel}} = 1 - F_{\text{parallel}} = 1 - 0.0133 = 0.9867$

2. Reliability of the other components in series:

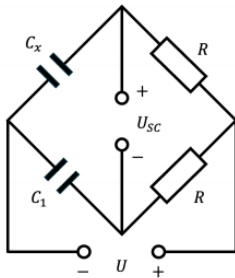
- Signal conditioning module: $R_{sc} = e^{-\lambda_{sc} t} = e^{-0.8 \times 1} \approx 0.4493$
- Data acquisition module: $R_{da} = e^{-\lambda_{da} t} = e^{-0.25 \times 1} \approx 0.7788$

3. Overall system reliability (R_{sys}):

$$R_{\text{sys}} = R_{\text{parallel}} \times R_{sc} \times R_{da}$$

$$R_{\text{sys}} = 0.9867 \times 0.4493 \times 0.7788 \approx 0.3453$$

4. A measurement system uses a mixed capacitive-resistive Wheatstone bridge as shown below:



One arm is a variable capacitive sensor C_x , another arm is a fixed capacitor $C_1 = \frac{2}{\pi} \times 10^4 \text{ pF}$. The other two arms are both resistors $R = 1 \text{ k}\Omega$. The AC excitation voltage amplitude is $U = 3 \text{ V}$, with a frequency of $f = 50 \text{ kHz}$. The bridge is initially balanced.

Note: The impedance of a capacitor is given by

$$Z_C = \frac{1}{j\omega C}$$

where $\omega = 2\pi f$.

a) Please Calculate the required capacitance C_x for the bridge to be balanced. (5 points)

b) If C_x decreases to $\frac{1}{\pi} \times 10^4 \text{ pF}$, calculate the effective (rms) voltage at the bridge output. (10 points)

Solution

Part a) Calculate the required capacitance C_x for the bridge to be balanced

1. Bridge Balance Condition:

For an AC Wheatstone bridge to be balanced (output voltage $U_{sc} = 0$), the ratio of the impedances in adjacent arms must be equal:

$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4}$$

Let the impedances of the four arms be:

- Top-left arm (C_x): $Z_1 = \frac{1}{j\omega C_x}$
- Bottom-left arm (C_1): $Z_2 = \frac{1}{j\omega C_1}$
- Top-right arm (R): $Z_3 = R$
- Bottom-right arm (R): $Z_4 = R$

2. Solve for C_x :

Substitute the impedance expressions into the balance equation:

$$\frac{\frac{1}{j\omega C_x}}{\frac{1}{j\omega C_1}} = \frac{R}{R} \quad \frac{C_1}{C_x} = 1 \quad C_x = C_1$$

Given $C_1 = \frac{2}{\pi} \times 10^4$ pF, we have:

$$C_x = \frac{2}{\pi} \times 10^4 \text{ pF}$$

Solution

Part b) Calculate the effective (rms) voltage at the bridge output if C_x decreases to $\frac{1}{\pi} \times 10^4$ pF

1. Calculate the angular frequency ω :

Given $f = 50$ kHz $= 50 \times 10^3$ Hz:

$$\omega = 2\pi f = 2\pi \times (50 \times 10^3) = 10^5 \pi \text{ rad/s}$$

2. Calculate the impedances of the capacitors:

Given $R = 1$ k $\Omega = 1000$ Ω .

○ Impedance of C_x (Z_{C_x}):

$$C_x = \frac{1}{\pi} \times 10^4 \text{ pF} = \frac{1}{\pi} \times 10^{-8} \text{ F}$$

$$Z_{C_x} = \frac{1}{j\omega C_x} = \frac{1}{j(10^5 \pi) \times (\frac{1}{\pi} \times 10^{-8})} = \frac{1}{j \times 10^{-3}} = -j1000 \Omega$$

○ Impedance of C_1 (Z_{C_1}):

$$C_1 = \frac{2}{\pi} \times 10^4 \text{ pF} = \frac{2}{\pi} \times 10^{-8} \text{ F}$$

$$Z_{C_1} = \frac{1}{j\omega C_1} = \frac{1}{j(10^5 \pi) \times (\frac{2}{\pi} \times 10^{-8})} = \frac{1}{j \times 2 \times 10^{-3}} = -j500 \Omega$$

3. Calculate the node voltages:

Treating the left node as the 0V reference and the right node as the excitation voltage $U = 3$ V, use the voltage divider rule to find the potentials at the top and bottom nodes:

○ Top node voltage (V_{top}):

$$V_{top} = U \frac{Z_{C_x}}{Z_{C_x} + R} = 3 \frac{-j1000}{1000 - j1000} = 3 \frac{-j}{1 - j} = 3 \frac{-j(1 + j)}{1^2 + (-1)^2} = 1.5 - j1.5 \text{ V}$$

○ Bottom node voltage (V_{bot}):

$$V_{bot} = U \frac{Z_{C_1}}{Z_{C_1} + R} = 3 \frac{-j500}{1000 - j500} = 3 \frac{-j0.5}{1 - j0.5} = 3 \frac{-j}{2 - j} = 3 \frac{-j(2 + j)}{2^2 + (-1)^2} = 0.6 - j1.2 \text{ V}$$

Solution

4. Calculate the output voltage amplitude:

The output voltage is the difference between the top and bottom nodes:

$$U_{sc} = V_{top} - V_{bot} = (1.5 - j1.5) - (0.6 - j1.2) = 0.9 - j0.3 \text{ V}$$

The amplitude (magnitude) of U_{sc} is:

$$|U_{sc}| = \sqrt{0.9^2 + (-0.3)^2} = \sqrt{0.81 + 0.09} = \sqrt{0.9} \approx 0.9487 \text{ V}$$

5. Calculate the effective (rms) voltage:

Since the input $U = 3\text{V}$ is the amplitude, the result above is also an amplitude. The rms voltage is the amplitude divided by $\sqrt{2}$:

$$U_{sc(rms)} = \frac{|U_{sc}|}{\sqrt{2}} = \frac{\sqrt{0.9}}{\sqrt{2}} = \sqrt{0.45} \approx 0.6708 \text{ V}$$

5. A moving coil indicator is connected to a Thévenin signal source of resistance $125\ \Omega$. Use the data given below to answer the following questions. The coil shape is square and the point angle θ is small.

Number of turns on coil = 150

Coil resistance = $25\ \Omega$

Coil area = $10^{-4}\ m^2$

Coil moment of inertia = $2.5 \times 10^{-5}\ kg\ m^2$

Magnetic flux density = $150\ Wb\ m^{-2}$

Spring stiffness = $10^{-3}\ Nm\ rad^{-1}$

- a) What is the sensitivity of the coil indicator? (5 points)
- b) What is the natural frequency for the system? (5 points)

Solution

Part a) Sensitivity of the coil indicator

For a moving coil indicator, the steady-state deflection occurs when the electromagnetic deflecting torque (T_d) equals the mechanical restoring torque (T_r) from the spring:

$$T_d = T_r$$
$$NBAI = K\theta$$

1. Calculate Current Sensitivity (S_I):

The current sensitivity (deflection per unit current) is:

$$S_I = \frac{\theta}{I} = \frac{NBA}{K}$$

Substitute the given values:

- $N = 150$ (Number of turns)
- $B = 150 \text{ Wb m}^{-2}$ (Magnetic flux density)
- $A = 10^{-4} \text{ m}^2$ (Coil area)
- $K = 10^{-3} \text{ Nm rad}^{-1}$ (Spring stiffness)

$$S_I = \frac{150 \times 150 \times 10^{-4}}{10^{-3}} = \frac{2.25}{10^{-3}} = 2250 \text{ rad/A}$$

2. Calculate System Voltage Sensitivity (S_V):

Since the indicator is connected to a Thévenin signal source, we must account for the total resistance in the circuit to find the overall voltage sensitivity ($\frac{\theta}{V}$).

Total resistance (R_{total}) = Source resistance (R_s) + Coil resistance (R_c):

$$R_{\text{total}} = 125 + 25 = 150 \Omega$$

Using Ohm's law ($I = \frac{V}{R_{\text{total}}}$):

$$S_V = \frac{\theta}{V} = \frac{S_I}{R_{\text{total}}}$$

$$S_V = \frac{2250}{150} = 15 \text{ rad/V}$$

Solution

Part b) Natural frequency for the system

The moving coil indicator acts as a second-order torsional mechanical system. The undamped natural angular frequency (ω_n) depends only on the spring stiffness (K) and the moment of inertia (J).

1. Calculate Natural Angular Frequency (ω_n):

$$\omega_n = \sqrt{\frac{K}{J}}$$

Substitute the given values:

- $K = 10^{-3} \text{ Nm rad}^{-1}$

- $J = 2.5 \times 10^{-5} \text{ kg m}^2$

$$\omega_n = \sqrt{\frac{10^{-3}}{2.5 \times 10^{-5}}} = \sqrt{\frac{100 \times 10^{-5}}{2.5 \times 10^{-5}}} = \sqrt{40} \approx 6.32 \text{ rad/s}$$

2. Calculate Natural Frequency in Hertz (f_n):

$$f_n = \frac{\omega_n}{2\pi} = \frac{\sqrt{40}}{2\pi} \approx \frac{6.3246}{6.2832} \approx 1.01 \text{ Hz}$$

6. An ideal displacement sensor can be described by $V = KI + a = 0.02I + 0.15$, at 22 °C and with power supply voltage 4.5 V. Here, V is the voltage output in Volts, I is displacement in meters, and a is the bias in Volts.

However, the power supply voltage (I_I) and environment temperature (I_M) will both affect the output. The comprehensive model is:

$$O = KI + a + K_M I_M I + K_I I_I$$

Where $K_M = 0.0001 \frac{V}{m^{\circ}C}$ and $K_I = 0.008 \frac{V}{V}$.

- If the sensor is used in a 32 °C environment and the power supply is 5.2 V, what are the new sensitivity and bias? (5 points)
- Suppose the sensor measures a displacement of 0.35 m in the environment above, What is the percentage error between the comprehensive model and ideal model? (5 points)

Solution

Part a) Calculate the new sensitivity and bias

The comprehensive model for the sensor is given as:

$$O = KI + a + K_M I_M I + K_I I_I$$

We can rearrange this equation to group the terms with the input displacement I (sensitivity) and the constant terms (bias):

$$O = (K + K_M I_M)I + (a + K_I I_I)$$

From this, we can define:

- **New Sensitivity:** $K_{\text{new}} = K + K_M I_M$
- **New Bias:** $a_{\text{new}} = a + K_I I_I$

1. Calculate the deviations (modifying and interfering inputs):

The deviations I_M and I_I are the differences between the new environment and the calibration (ideal) environment.

- Calibration conditions: 22°C , 4.5 V
- New conditions: 32°C , 5.2 V
- Temperature deviation (I_M): $I_M = 32 - 22 = 10^\circ\text{C}$
- Voltage deviation (I_I): $I_I = 5.2 - 4.5 = 0.7\text{ V}$

2. Calculate New Sensitivity (K_{new}):

$$K_{\text{new}} = 0.02 + 0.0001 \times 10 \quad K_{\text{new}} = 0.02 + 0.001 = 0.021\text{ V/m}$$

3. Calculate New Bias (a_{new}):

$$a_{\text{new}} = 0.15 + 0.008 \times 0.7$$

$$a_{\text{new}} = 0.15 + 0.0056 = 0.1556\text{ V}$$

Solution

Part b) Calculate the percentage error between the comprehensive model and ideal model

1. Calculate the ideal output (V_{ideal}):

Using the ideal model $V = 0.02I + 0.15$ with a displacement $I = 0.35$ m:

$$\begin{aligned}V_{\text{ideal}} &= 0.02(0.35) + 0.15 \\V_{\text{ideal}} &= 0.007 + 0.15 = 0.157 \text{ V}\end{aligned}$$

2. Calculate the comprehensive output (O_{comp}):

Using the new sensitivity and bias from Part a for $I = 0.35$ m:

$$\begin{aligned}O_{\text{comp}} &= 0.021(0.35) + 0.1556 \\O_{\text{comp}} &= 0.00735 + 0.1556 = 0.16295 \text{ V}\end{aligned}$$

3. Calculate the percentage error:

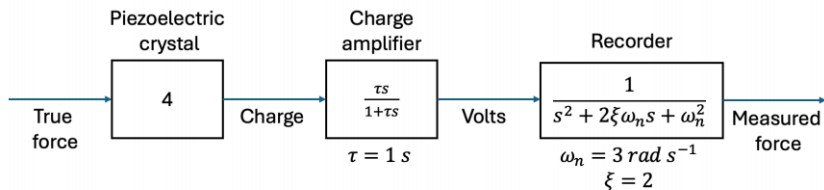
The percentage error is calculated relative to the ideal (expected) true value:

$$\text{Percentage Error} = \frac{O_{\text{comp}} - V_{\text{ideal}}}{V_{\text{ideal}}} \times 100\%$$

$$\text{Percentage Error} = \frac{0.16295 - 0.157}{0.157} \times 100\%$$

$$\text{Percentage Error} = \frac{0.00595}{0.157} \times 100\% \approx 3.7898\%$$

7. A force measurement system consisting of a piezoelectric crystal, charge amplifier, and recorder is shown in the figure below.



Calculate the system dynamic error function corresponding to the force input signal: (15 points)

$$F(t) = 5(\sin 1t + \frac{1}{2}\sin 2t)$$

Solution

Calculate the system dynamic error function

The overall system transfer function $H(s)$ is the product of the transfer functions of the three components in series:

$$H(s) = 4 \times \left(\frac{s}{s+1} \right) \times \left(\frac{1}{s^2 + 12s + 9} \right)$$

Substituting $s = j\omega$ to get the frequency response function:

$$H(j\omega) = \frac{j4\omega}{(1+j\omega)[(9-\omega^2) + j12\omega]}$$

From this, we define the general formulas for the amplitude ratio $|H(j\omega)|$ and phase shift $\angle H(j\omega)$

- Amplitude ratio: $|H(j\omega)| = \frac{4\omega}{\sqrt{1^2 + \omega^2} \cdot \sqrt{(9-\omega^2)^2 + (12\omega)^2}}$
- Phase shift: $\angle H(j\omega) = 90^\circ - \arctan(\omega) - \arctan\left(\frac{12\omega}{9-\omega^2}\right)$

1. For the $\omega_1 = 1 \text{ rad s}^{-1}$ component (Input amplitude $A_1 = 5$):

- Amplitude ratio:

$$|H(j1)| = \frac{4(1)}{\sqrt{1^2 + 1^2} \cdot \sqrt{8^2 + 12^2}} = \frac{4}{\sqrt{2} \cdot \sqrt{208}} = \frac{4}{\sqrt{416}} \approx 0.1961$$

- Phase shift:

$$\angle H(j1) = 90^\circ - \arctan(1) - \arctan\left(\frac{12}{8}\right) = 90^\circ - 45^\circ - 56.31^\circ = -11.31^\circ$$

Solution

2. For the $\omega_2 = 2 \text{ rad s}^{-1}$ component (Input amplitude $A_2 = 2.5$):

○ Amplitude ratio:

$$|H(j2)| = \frac{4(2)}{\sqrt{1^2 + 2^2} \cdot \sqrt{5^2 + 24^2}} = \frac{8}{\sqrt{5} \cdot \sqrt{601}} = \frac{8}{\sqrt{3005}} \approx 0.1459$$

○ Phase shift:

$$\angle H(j2) = 90^\circ - \arctan(2) - \arctan\left(\frac{24}{5}\right) = 90^\circ - 63.43^\circ - 78.23^\circ = -51.66^\circ$$

○ Steady-state output component $O_2(t)$:

$$O_2(t) = 2.5 \times 0.1459 \sin(2t - 51.66^\circ) = 0.3648 \sin(2t - 51.66^\circ)$$

3. Calculate System Dynamic Error Function $E(t)$:

The dynamic error is defined as the actual measured output minus the true input signal:

$$E(t) = [O_1(t) + O_2(t)] - F(t)$$

Group the corresponding frequency terms together:

$$E(t) = [O_1(t) - F_1(t)] + [O_2(t) - F_2(t)]$$

Answer: The system dynamic error function is:

$$E(t) = [0.9805 \sin(t - 11.31^\circ) - 5 \sin(t)] + [0.3648 \sin(2t - 51.66^\circ) - 2.5 \sin(2t)]$$